ANUK PUBLIC SECTOR ACCOUNTING & FINANCE JOURNAL

Contact Address:

ANAN UNIVERSITY BUSINESS SCHOOL,
14 Road, Off 1st Avenue,
Gwarimpa, Abuja

September, 2025

Published September 2025

Web Address: hyyps://www.anukpsaj.com Email: anukpsafj@gmail.com

Printed by:
EDUCODEX LTD
5 Otukpo Street, Area 11 Garki, Abuja.
08055006444, 08148134773
Email: educodex1@gmail.com
Website: www.educodex.com

EDITORIAL

It is with great pleasure and a deep sense of academic fulfillment that we present the maiden edition of the Public Sector Accounting and Finance Journal (PSAFJ), published by ANAN University, Kwall. This inaugural issue marks a significant milestone in the University's commitment to advancing scholarship, policy, and practice in the field of public sector accounting, finance, and governance.

In an era when nations are striving for transparency, accountability, and fiscal sustainability, the role of public sector accounting and finance has never been more critical. The journal is therefore conceived as a platform for intellectual engagement, rigorous research, and policy discourse that contribute meaningfully to the strengthening of public financial management systems, good governance, and sustainable development in Nigeria and beyond.

This maiden edition features ten scholarly articles drawn from a wide spectrum of research interests and methodologies within the field. The papers collectively address critical issues ranging from public sector financial reporting and accountability, budgeting reforms, internal control mechanisms, audit quality, fiscal transparency, to the application of emerging technologies in public finance management. Each contribution reflects the authors' depth of research and commitment to the continuous improvement of public sector governance and accountability.

COLLEGE OF PUBLIC SECTOR ACCOUNTING
ANUK Public Sector Accounting and Finance Journal (APSAFJ)

THE OBJECTIVE OF THE JOURNAL

To provide a platform for researchers and academics to publish research papers and articles within the research domains of public sector accounting, public finance, public economics, environmental, oil and gas accounting, public financial management, taxation and fiscal policy, public policy and finance, macroeconomics and development, public treasury management, and other related areas..

ETHICS AND POLICIES

1. Publication Ethics
At the ANUK Public Sector Accounting and Finance Journal (APSAFJ), we uphold the highest standards of publication ethics to ensure the integrity and quality of our scholarly outputs. We are committed to ethical publishing practices, guided by the principles of the **Committee on Publication Ethics (COPE)**. All authors, reviewers, and editors are expected to adhere to the following ethical standards:

- **Plagiarism:** All submitted manuscripts undergo plagiarism detection using [Plagiarism Detection Tool, e.g., Turnitin]. Manuscripts found to contain significant plagiarism, including self-plagiarism, will be rejected. Authors are encouraged to properly cite and attribute all sources.
- **Authorship:** Only individuals who have significantly contributed to the research and manuscript preparation should be listed as authors. Ghostwriting, guest authorship, and other forms of unethical authorship are strictly prohibited.
- **Research Misconduct:** Fabrication, falsification, or misrepresentation of data are unacceptable. Any suspected cases of research misconduct will be investigated and may lead to retraction of the published work.

2. Conflict of Interest Disclosures
To maintain transparency and objectivity, all authors, reviewers, and editors must disclose any potential conflicts of interest that could influence the outcome or interpretation of their work. Conflicts of interest may include, but are not limited to:

- Financial relationships (e.g., funding, employment, stock ownership).
- Personal relationships or competing interests that could affect the impartiality of the research.

Authors must include a **Conflict of Interest Statement** in their manuscript, even if no conflicts exist (e.g., "The authors declare no conflicts of interest"). Reviewers and editors must recuse themselves from handling manuscripts where conflicts of interest exist.

3. Data Availability and Sharing Policies
To promote transparency and reproducibility in research, ANUK Public Sector Accounting and Finance Journal (APSAFJ) encourages authors to share their data wherever possible. Our data sharing policy is as follows:

- **Data Availability Statement:** Authors must include a statement in their manuscript specifying whether and how the data supporting their findings are available. Examples include:
    - "The data supporting this study are available in [repository name, DOI or link]."
    - "The data used in this study are available upon reasonable request from the corresponding author."
- **Preferred Repositories:** Authors are encouraged to deposit their data in trusted repositories that provide DOIs or persistent identifiers (e.g., Dryad, Figshare).
- **Ethical Considerations:** Data involving human participants must comply with ethical guidelines and institutional requirements, including anonymization where necessary.
- **Open Data:** Where possible, authors should consider licensing their data under open terms (e.g., Creative Commons licenses) to facilitate reuse.

EDITORIAL POLICIES

1. Peer Review Process
At the ANUK Public Sector Accounting and Finance Journal (APSAFJ), we employ a rigorous peer review process to ensure the highest standards of scholarly integrity and quality in published research. Our peer review system is described below:

- **Type of Review:** We use a **double-blind peer review process**, where:
    - In **double-blind review**, both the authors and reviewers remain anonymous to each other to ensure impartiality.
- **Reviewer Selection:** Reviewers are selected based on their expertise in the subject matter of the manuscript.
- **Timeline:** The peer review process typically takes 10 – 20 days, depending on the complexity of the manuscript and reviewer availability.
- **Decision Criteria:** Editorial decisions are based on originality, methodological rigor, significance to the field, and adherence to ethical standards.

2. Publication Frequency

- **Frequency of Publication:** ANUK Public Sector Accounting and Finance Journal (APSAFJ) publishes on a **quarterly** basis, with issues released in June and December.
- **Special Issues:** In addition to regular issues, the journal occasionally publishes special issues on emerging topics of interest.

3. Copyright and Licensing

- **Copyright Policy:** APSAJ retains the copyright to the article submitted to the journal, the rights to publish and distribute their articles. The authors' rights include:
    - Reuse and redistribution of the published work.
    - Use of the work in academic and educational settings.
- **Licensing:** Articles are published under a **Creative Commons Attribution License (CC BY)**, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.
    - License Details: Link to the specific license, e.g., CC BY 4.0.

4. Archiving Policy
To ensure long-term preservation and access to published content, ANUK Public Sector Accounting and Finance Journal (APSAFJ) has implemented robust archiving strategies, including:

- **LOCKSS (Lots of Copies Keep Stuff Safe):** A decentralized preservation system that ensures content remains accessible even in the event of a system failure.
- **CLOCKSS (Controlled LOCKSS):** Content is preserved in trusted digital archives for secure, long-term storage.
- **Institutional Repositories:** Articles are deposited in [specific repositories or institutional archives], ensuring perpetual access.
- **Self-Archiving Policy:** Authors are encouraged to deposit the published version of their articles in institutional or subject-specific repositories.

Our commitment to archiving ensures that your research remains accessible and discoverable for future generations.

5. Standard Journal Content Format

The standard format for content is as follows;

1. Introduction
2. Statement of Research Problem
3. Research Hypotheses or Objectives
*Review of Related Literature
4. Data Presentation & Analysis
5. Research Findings & Discussion of Findings
6. Conclusion, Recommendation & Policy Implication
7. References
8. Appendices

Or

1. **Indroduction** (Background of the study, research gap, research hypotheses, scope of study, significance of the study).

2.0. Literature Review
2.1 Conceptual Review
2.2 Empirical Review
2.3 Theoritical Review’

3.0 Methodology
3.1 Research Design
3.2 Population sample size and sampling techniques
3.3 Analytical Tools
3.4 Model Specification

4.0 Data Presentation & Analysis
4.1 Data Presentation
4.2 Analysis & Findings
4.3 Discussion of Findings

5.0 Conclusion
5.1 Recommendations & Policy Implication

References

Appedices

TABLE OF CONTENTS

TAX POLICY AND FISCAL SUSTAINABILITY IN NIGERIA

IBRAHIM LUKA TUMBA

ibroltt.il@gmail.com

DEPARTMENT OF TAXATION AND FISCAL POLICY

ANAN University Kwall, Plateau State, Nigeria

Abstract

Tax policy is one of the primary tools that governments use to achieve fiscal sustainability and economic growth. In Nigeria, there have been various tax policies aimed at increasing revenue generation, promoting equity, and enhancing the country's economic stability. This study assesses the impact of tax policies on the fiscal sustainability of the Nigerian state. The study employs the use of secondary data. Tax policy in this study is proxied with Petroleum Profit Tax, Company Income Tax and Value Added Tax. Fiscal sustainability is denoted as government expenditure and revenue measured as a ratio of Gross Domestic Product (GDP). The study deployed an ex-post facto research design while the Vector Error Correction Model was used to analyse the data gathered. Findings of the study revealed that the three variables significantly influence fiscal sustainability. However, company income tax was negatively significant, while petroleum profit tax and value added tax were positively significant. Based on the findings, the study recommends, among others, that trainings, the papers, and workshops should be organized by government tax agencies to the Nigerian public, students in higher institutions, and companies on the importance and benefits of tax revenue to the economy.

Keywords: Tax policy, petroleum profit tax, company income tax, value added tax, fiscal sustainability

1. INTRODUCTION

Taxation is a crucial tool for generating revenue a promoting economic development in any nation. I Nigeria, tax revenue accounts for an average of 6% Gross Domestic Product (GDP). This low percenta demonstrates the inadequacy and inefficiency Nigeria's tax system. Nigeria is dependent on exports for revenue, which exposes the country to t risks arising from global oil price fluctuations. T importance of taxation to a country's socio-econon development cannot be overemphasized. Taxation one of the important sources of revenue governments to fund essential public servic improve infrastructure, and promote econon activities. In Nigeria, taxation policies have be adapted to aid the government in the mobilization funds for infrastructure development, pove reduction, and other essential public services. Desp these efforts, Nigeria's economy continues to be at r due to a lack of fiscal sustainability. The tax system Nigeria is primarily a mix of direct and indirect ta arranged in the form of Personal Income Tax (PI Petroleum profit tax, Company Income Tax (CI Value Added Tax (VAT), and Custom Excise Tari (CET). The Federal Inland Revenue Service (FIR and State Internal Revenue Service (SIRS) a responsible for administering and implementing t tax policies in Nigeria. Nigeria's tax revenue inadequate, particularly when compared to its W African neighbours, Ghana, and Kenya. The countr Gross Domestic Product (GDP) to tax ratio is only 6%, compared to the average ratio of 21% of African nations (Ndikom, 2020). Direct taxes such as PIT and CIT are levied on individuals and corporations, respectively, while VAT and CET are indirect taxes. VAT remains the major contributor to Nigeria's total tax revenue, accounting for 51.8% in 2020. VAT is regarded as a consumption tax and is levied on goods and services consumed in Nigeria (PWC, 2020). It is believed that a sound tax policy will have implications for fiscal sustainability.

Sustainability of fiscal policy is essentially a macroeconomic concept which is related to the solvency of public's coffers. Solvency occurs if current public fiscal obligations do not lurk future responsibility of government. Fiscal policy is said to be sustainable when an economy is anticipated to finance its debts deprived of large imminent modifications to equilibrium of revenue and expenditure. Alternatively, sustainability denotes that government finances are not resorting to debts monetization or repudiation and a sizable level of external shock is unanticipated to bring the economy into unending debts (Kojo, 2010; Kaur et. al., 2018; Chibi et. al., 2019). Issue of fiscal sustainability is central to measuring macroeconomic soundness. A sustainable policy provides steadiness for the macroeconomic environment and promotes monetary union relationships among economies Oyeleke and Ajilore (2014). It also encourages growth and development of an economy. On the other hand, unsustainable fiscal policy threatens macroeconon strength, balance of payment equilibrium and financ aptitude of government to supply indispensable goc and services. When fiscal strength of government apparently unsustainable over time, the markets fr which the government finances its debts react triggering macroeconomic crises (Sudharshan et a 2012). Furthermore, unsustainable fiscal policy, in t long run, would jeopardize efforts of the monet authority at attaining price stability. Operationally, expansion in the money supply will increase the le of importation, thereby dipping the aggregate foreign exchange available to domestic consumers a swelling pressure for exchange rate devaluation. In t literature, sustainable fiscal policy is augmenting t soundness of macroeconomic environment promoting growth of economic and fiscal discipline is therefore central for policy makers, especially Nigeria to focus attention on its attainment as a poli measure of good economic management.

A comprehensive tax education campaign is a necessary to create awareness and educate taxpay on the importance of paying taxes. To increa revenue generation, Nigeria must broaden its tax ba This can be achieved by reducing tax exemptions tl target specific individuals or companies a expanding the tax net to capture other revenu generating activities. The government should a explore the possibilities of imposing a tax on digi transactions to generate additional reven Furthermore, implementing efficient t administration practices can also improve reven generation and tax compliance. Automating the t collection process and ensuring a seamless t payment system will enhance revenue mobilizati and reduce corruption. Nigeria should also establisl strong anti-graft institution to prevent reven leakages and enforce tax laws. Therefore, this stu seeks to investigate the effect of tax policy on t fiscal sustainability of states Nigeria.

2. STATEMENT OF THE PROBLEM

Despite the implementation of a range of tax polic in Nigeria, the country grapples with an enduring a substantial fiscal deficit. This ongoing challenge l spawned a cascade of detrimental consequenc encompassing critical issues such as an acute shorta of resources for vital developmental programs, deteriorating infrastructure base, an eleva prevalence of poverty, and alarming levels unemployment. The magnitude of this fiscal defi problem underscores the urgent need for comprehensive examination of the effectiveness a implications of Nigeria's taxation strategies. Theref it is important to investigate the effectiveness of t tax policies in Nigeria and to determine their imp on the fiscal sustainability of the Nigerian state. T research hence aims to fill this gap by examining the relationship between tax policies and fiscal sustainability in Nigeria.

3. HYPOTHESES

For clarity, three hypotheses were formulated to provide direction for the study. These hypotheses are stated in null form.

Hypothesis One: Petroleum profit tax has no significant effect on fiscal sustainability in Nigeria

Hypothesis Two: Company income tax has no significant effect fiscal sustainability in Nigeria

Hypothesis Three: Value added tax has no significant effect on fiscal sustainability in Nigeria.

4. LITERATURE REVIEW

4.1. Conceptual Review

According to the report of the presidential committee on National Tax policy (2008), "The National tax policy provides a set of rules, modus operandi and guidance to which all stakeholders in the tax system must subscribe". Tax policy formulation in Nigeria is the responsibility of the Federal inland Revenue Services (FIRS), Customs, Nigerian National Petroleum Corporation (NNPC), National Population Commission (NPC), and other agencies but under the guidance of the National Assembly i.e., the law-making body in Nigeria (Presidential committee on National tax policy, 2008). Suffice it to say that if there must be any effective implementation of the Nigerian tax system or attainment of its goal, the use of the national tax policy document remains essential. According to the Presidential Committee on tax policy (2008), "Nigeria needs a tax policy which does not only describe the set of guiding rules and principles, but also provide a stable point of reference for all the stakeholders in the country and upon which they can be held accountable. James and Nobes (2008) decried the inability of tax policy to meet up with efficiency and equity criteria against which it is being judged. It was further noted that tax policy is continually subjected to pressure and changes which most time does not guarantee outcome that are in line with the overall goal (James and Nobes, 2008). Unfortunately, most policy changes in Nigeria are without adequate consideration of the taxpayers, administrative arrangement and cost plus the existing taxes. This has in no small measure hindered the effective implementation and goal congruence of the nation's tax system. Citing (Bird and Oldman,1990), James and Nobes (2008) stated as follows "the best approach to reforming taxes is one that takes into account taxation theory, empirical evidence and political and administrative realities and blend them with good dose of local knowledge and a sound appraisal of t current macroeconomics and international situation produce a feasible set of proposals sufficien attractive to be implemented and sufficiently robust withstand changing times, with reason and s produce beneficial results".

The literature on taxation in Nigeria shows that th are several challenges that tax policy makers fa These challenges include weak administration syste corruption, unclear tax policies, inadequate pub awareness and limited tax base. Notwithstanding th challenges, several studies have reported that taxati can lead to revenue generation, poverty reduction, a economic growth (Ezepue, 2016; Taiwo and Oyinl 2018). This study also observes that tax policies a critical for a country's fiscal sustainability and t they can be used to address income inequality redistributing income and wealth among citizens. study by Akinboade and Fadare (2017) reveals tha well-planned tax system that eliminates unjustifi exemptions and broadens the tax base is essential fiscal sustainability. However, it is important highlight that tax policies should not only focus revenue generation, but also on promoting econon growth, as high tax rates can discourage investm and stifle economic activities.

4.2. Empirical Review and Study Gap

Several studies have explored tax policy and impact on the sustainability of states in Nigeria. study by Ndongo, Tanoh, and Yapo (2020) examin the impact of tax policy on economic and soc sustainability in Nigeria. The study found that t policy has a positive impact on economic grow poverty reduction, and social development, which a essential for the overall sustainability of states.

Another study by Akpan, Oyati, and Asamu (201 investigated the impact of tax policy on environmen sustainability in Nigeria. The study found that t policy could be an effective tool for promoti environmental sustainability in Nigeria. The stu recommended that tax policy should be designed promote green energy, reduce pollution, and conse biodiversity.

A study by Idowu et al. (2018) also explored t impact of tax policy on income inequality in Nige The study found that progressive tax policies co help to reduce income inequality and prom economic sustainability in Nigeria. The stu recommended that policymakers should design t policies that are progressive and equitable.

Ogunlana and Afolabi (2019) analysed t relationship between taxation and financ sustainability of Nigerian states between 1981 a 2016. They found that weak tax compliance a administration were major obstacles to the financ sustainability of Nigerian states. The auth recommend improvements in tax administration to enhance the revenue and financial stability of Nigerian states.

Adewuyi (2018) investigated the impact of tax policy on economic development using Oyo State as a case study. The study found a positive relationship between tax policy and economic development and suggests that a well-executed tax policy could be an effective tool for promoting economic growth and enhancing financial sustainability.

Iweala and Onyekwena (2020) analysed the relationship between tax policy, revenue generation, and financial sustainability of Nigerian states using a panel dataset. They found that a well-structured tax policy can significantly increase revenue and enhance financial sustainability of Nigerian states, and the authors recommended improvements in tax policies to promote economic growth and financial resilience.

Adewale and Adetiloye (2021) examined the relationship between taxation and fiscal sustainability using Lagos State as a case study. The study found that effective tax administration is vital for boosting revenue and financial sustainability of Lagos State, and they recommended improving tax policy and strategies to promote sustainable financial growth.

Akinola and Adedokun (2019) investigated the relationship between taxation, sustainable development, and resource mobilization in Nigeria. The study suggests that effective tax policies could promote sustainable development and enhance financial sustainability of Nigerian states. The authors recommend adopting progressive tax policies and improving tax administration to promote sustainable development.

While existing research has extensively explored the effects of taxation on fiscal sustainability in various contexts, there is a notable gap in the literature concerning the nuanced impact of several tax components on fiscal sustainability. Limited attention has been given to the ways in which some sectors may undermine government revenue collection efforts and subsequently affect fiscal sustainability in developing countries like Nigeria. Understanding the complex dynamics between tax compliance and fiscal sustainability is crucial for policymakers seeking to design effective taxation policies that bolster revenue streams and enhance fiscal stability in such contexts.

4.3. Theoretical Review

The Laffer curve theory is apt in explaining the relationship between taxation and fiscal sustainability. Professor Arthur Laffer proposed the Laffer curve theory in 1974 which explicates the theoretical relationship between the tax rate and government revenue derived from taxation. Laffer demonstrated the idea that a change in tax rate will have an effect on tax revenue in two different ways which are t Arithmetic Effect and the Economic Effect (Laff 2004). The Arithmetic Effect states that a depletion tax rate will bring about a cutback in tax revenue (p currency of the tax base) proportionate to t reduction in tax rate, and vice versa. The Econon Effect, on the other hand, recognizes that a lower t rate will have a favourable influence on work, outp employment, and, as a result, the tax base, by helpi to grow activities through incentives. The Arithme Effect is the polar opposite of the Economic Effect. a result, when the Economic and Arithmetic Effects tax rate transposes are combined, the impact of change in tax rates on total tax collection is l apparent.

Furthermore, the "benefit theory of taxation" relevant when it comes to tax policy and its impact the sustainability of states in Nigeria. This theo states that individuals and businesses should pay tax based on the benefits they receive from public goo and services (Oseni & Okwu, 2017). This implies tl those who benefit most from public goods a services should pay a higher tax rate than those w benefit less. This theory suggests that tax poli should be designed to promote equity, fairness, a sustainability.

5. METHODOLOGY

An ex-post facto research design was deemed suital for this paper because it encompasses a technique which concepts or categories with an already exta feature or element have been grouped and can compared to a specific dependent variable. For tl research the independent variable, taxation is divid into groups which are petroleum profit tax (PP' Company Income Tax (CIT), Value Added Tax (VA' and it will be regressed on the dependent variab fiscal sustainability (FS), measured by annual tii series of government expenditure and reven measured as a ratio of Gross Domestic Product (GD The location of this study is Nigeria. This resea covers the time period from 2000 to 2021. This stu made use of secondary data. This secondary data obtained from the Central Bank of Nigeria Statisti Bulletin and Federal Inland Revenue Service (FIR Data on Government expenditure, Governme revenue and Gross Domestic Product was obtain from the CBN Statistical Bulletin and the tax reven data is derived from published statements of t Federal Inland Revenue Service. The justification the choice of secondary data is that the informati can be obtained from published reports, with considerable measure of validity.

5.1. Method of Data Analysis

The data gathered for this research was subjected some form of analysis, employing a time series da which is a sequence of data collected over time. The Vector Error Correction Model was the analytical technique employed in this research study. The VECM model was used because the cointegration existed between the variables after the Johansen Cointegration test. A Johansen Cointegration test was used to check whether or not cointegration existed. Unit root test was also employed in this study to so as to test for stationarity in the time series. It is to determine whether the variables are stationary or non-stationary. Descriptive analysis was also performed because it is a useful tool for assessing and summarizing vast amounts of raw data. The descriptive analysis focuses on displaying data sets in their most basic form to convey fundamental information about variables in the dataset. Correlation analysis was also used to see if there were any plausible links between the variables. It is a statistical method for determining the strength of a link between variables. E-views 9 was used as the statistical package to perform the relevant data analysis.

5.2. Model Specification

Fiscal sustainability is the dependent variable for the study. The Linear specification is given as;

$$FS = f(PPT, CIT, VAT) \text{ ———————————— } (1)$$

From the above linear specification, the statistical model was derived. This is stated as

$$FS = \beta_1 PPT_t + \beta_2 CIT_t + \beta_3 VAT_t \text{ ————————— } (2)$$

The linear function was changed into log form for better interpretation of results. Therefore;

$$logFS_{it} = \alpha + \beta_1 logPPT_{it} + \beta_2 logCIT_{it} + \beta_3 logVAT_{it} + \mu_{it} \text{ ———————— } (3)$$

Where:

FS = Fiscal sustainability

PPT = Petroleum Profit Tax

CIT = Company Income Tax

VAT = Value Added Tax

α is a constant

β_1, β_2, β_3 are the coefficients of the parameter estimate

μ is the error term.

6. DATA PRESENTATION AND ANALYSIS

This chapter deals with the analysis and presentation of data as well as the interpretation of results. The econometric techniques used in the study to determine the effect of taxation on fiscal sustainability in Nigeria were the Augmented Dickey-Fuller unit root test, Johansen Cointegration test, and Vector Error Correction Model analysis.

Table 1. Measurement of Variables		
VARIABLE NAME	**VARIABLE MEANING**	**VARIABLE MEASUREMENT**
DEPENDENT VARAIABLE		
FS	Fiscal Sustainability	Annual time series of government expenditure and revenue measured as a ratio of Gross Domestic Product (GDP).
INDEPENDENT VARIABLES		
PPT	Petroleum Profit Tax (now referred to as Hydrocarbon Tax following the implementation of Petroleum Industry Act (PIA) 2021.	50% for oil and gas operations under production sharing agreements (PSC)N on-PSC operations, including joint ventures (JVs) are charged 65.75 percent in the first five years and 85 percent after the first five years. Profits from upstream gas are taxed at 30%.
CIT	Company Income Tax	30% tax for large companies and is charged on profits for the accounting year ending in the year preceding assessment.
VAT	Value Added Tax	7.5% tax is charged on goods provided in Nigeria or imported into Nigeria from the year 2020 till 2021 while 5% was charged from 2000–2019

Source: Researcher's computation

6.1 Descriptive Statistics

Table 2. Descriptive Analysis of Variables

	LFS	**LCIT**	**LPPT**	**LVAT**
Mean	10.82193	6.141480	7.138907	9.219214
Median	10.93387	6.486880	7.254766	7.485234
Maximum	11.18987	7.466222	8.071318	13.24439
Minimum	10.13339	3.975936	5.413430	6.176714
Std. Dev.	0.345601	1.112525	0.734138	2.736048

Skewness	−0.623309	−0.618258	−0.893227	0.233309
Kurtosis	2.458793	2.989934	2.944023	3.202724
Sum	238.0825	135.1126	157.0560	202.8227
Sum Sq. Dev.	2.508240	25.99197	11.31812	157.2052
Observations	22	22	22	22

Source: Researcher's computation

Table 2, which represents the Descriptive statist results showed the attributes and nature of the varial used in the study. It showed each variable's medi mean, minimum and maximum values, stand deviations, kurtosis, skewness, probability and t sum of square deviations. The described variab were log of fiscal sustainability (LFS) which is t dependent variable and, log of petroleum profit t (LPPT), log of company income tax (LCIT), and l of value added tax (LVAT) which are the independ variables.

Both the mean and the median served as indicators central tendency. The greatest values for the log (LFS) were 10.82193 and 10.93387, respectively. T greatest value for the log of value added tax (LVA was 13.24439, while the minimum value for the log company income tax (LCIT) was 3.975936. T standard deviation displayed the summation squared deviations from the mean and log of val added tax (LVAT) had the highest value of 2.7360 The skewness made the asymmetry of the d distributions around its mean apparent.

The skewness of the normal distribution typica ranges from 0 to 1. As a result, a positive skewn indicated that the data distribution has a long right t while a negative skewness suggested that t distribution has a long-left tail. Due to their range o to 1, all of the variables (LFS, LCIT, LPPT, and LV had a normal distribution skewness. Only LVAT w positively skewed, with a value of 0.233309, wher the variables (LFS, LCIT, and LPPT) were advers skewed due to their negative values of −0.6233 −0.618258, and −0.893227.

Kurtosis showed if the data series' spreads were flat peaks. The variable is considered to have peaked the normal if the kurtosis value is 3 or above, which excellent because it indicates that the variable uniformly distributed. The distribution is conside to be flat to the normal if the kurtosis is under 3. W kurtosis values of 2.989934, 2.944023, and 3.2027 respectively, the variables LCIT, LPPT, and LV were peaked to the normal distribution whereas t variable LFS is flat to the normal distribution becau its value of 2.458793 is below 3.

2 Correlation Analysis

The type of relationship between the independent variable, LFS and all the independent variables LCIT, LPPT, and LVAT were tested using this. A positive or negative relationship could exist. In contrast to a negative relationship, which denotes an indirect or inverse relationship between the two variables, a positive relationship indicates a direct association between the dependent and independent variables.

According to Table 2 the independent variables LPPT and LVAT and the dependent variable LFS had positive correlations with values of 0.499365 and 0.064288, respectively. Only LCIT, with a value of −0.477703, had a negative correlation with LFS. This demonstrated that the impact of the independent variables on the dependent variable was balanced.

3 Unit Root Test

This was used to test for the stationarity of the variables so as to ascertain their order of integration. Using the Augmented Dickey Fuller (ADF) test, the ADF unit root test ascertains the presence of a unit root (non-stationary) tested against alternative hypothesis of the absence of a unit root (stationary). The ADF statistic (in absolute figure terms) must be greater than the standard critical value at 5% level of significance.

The ADF result demonstrated that all variables were stationary at the first difference, trend, and intercept since the ADF statistic's value IS above the standard critical value at a significance level of 5%. The Johansen cointegration test was used to confirm the inference that the variables had a long-term relationship. The ADF test results are displayed in Table 4.

Table 3. Correlation Analysis Result				
	LFS	**LCIT**	**LPPT**	**LVAT**
LFS	1.000000	−0.477703	0.499365	0.064288
LCIT	−0.477703	1.000000	0.937139	0.179583
LPPT	0.499365	0.937139	1.000000	0.177517
LVAT	0.064288	0.179583	0.177517	1.000000

EViews output

Table 4. ADF unit root test results					
Variable	**ADF @ 1st Differen**	**5% Criticalvalu**	**Order of Integrati**	**Included inthe Moc**	**Remark**
LFS	−5.481635	−3.658446	I(1)	Trend and Intercept	Stationary
LCIT	−5.440434	−3.658446	I(1)	Trend and Intercept	Stationary
LPPT	−4.586165	−3.658446	I(1)	Trend and Intercept	Stationary
LVAT	−5.053377	−3.673616	I(1)	Trend and Intercept	Stationary

EViews Output

6.4 Johansen Co-Integration
The long-run relationship was estimated adopting t Johansen co-integration test to ascertain the long-r relationship among the variables in the equation. T null and alternate hypothesis is:

H_0: There is no long-run relationship among t variables in the model

H_1: There is a long-run relationship among t variables in the model

From the Johansen co-integration analysis in Table and using lag 1, there was a long-run relationsh between log of FS (LFS) which was the depend variable and log of company income tax (LCIT), l of petroleum profit tax (LPPT), and log of value adc tax (LVAT) which were the independent variabl From the unrestricted cointegration rank test; usi the trace statistics in Table 4, there existed four cointegrating equation, with the probability value l than 0.05 level of significance and figures of 0.00 0.0346, 0.0289, and 0.0097. The maximum Eig value on the other hand from Table 5 indicated a four (4) cointegrating equations with figures of 0.00 0.0363, 0.0185, and 0.0097. Therefore, the res implies that using both the trace statistics and t maximum Eigen value statistics, there was a long-r relationship in the model. This also means that the n hypothesis was rejected, meaning that there is a long-run relationship among the variables in the model. Since a long-run relationship existed, the vector error correction model (VECM) was conducted to determine the long-run and short-run association and thus the corresponding speed of adjustment. The result is thus represented in Table 4 and 5.

5 Vector Error Correction Model (VECM)
VECM is used to calculate the speed at which short disequilibrium converge into long-run equilibrium relationship. It is constructed only if the variables are co-integrated which implied that there was evidence of a long-run relationship among the variables. Moreover, it is a restricted VAR model with co-integrating restrictions imputed into its specification. The coefficients in the co-integrating equation showed the estimated long-run relationship among the variables. The coefficient in the VECM revealed how deviations from that long-run relationship affect the changes in the variable in the next time- period. In summary, those coefficients across cointEq1 under the VECM show how each variable will move in the next period to get back to the long-run relationship. To confirm the Johansen co- integration conclusion, Table 5 directs us to first look at the coefficient values of the cointeq1 to determine whether there is a long-term

relationship among the model's variables. T coefficient values of each independent variable a then scrutinized to determine whether they w quickly resume their long-term relationships with t dependent variable in the upcoming period.

From Table 5, the first figure under each variable w the coefficient. The second values in circle brack represented the standard error, while the third value square brackets represented the t-statistics. T cointeq1 value of −0.746375 indicated a long-r relationship between the independent variables and t dependent variable LFS, representing a hi percentage of about 75%, which is near to 100%. T next step is to determine how quickly the long-r relationship can converge in the following time fran The coefficient value of log of company income t (LCIT) was −0.140119 to prove that it has a short-r time period in having a long-run impact on t dependent variable log of FS (LFS) both now and the next period. The short-run speed of adjustme was approximately 14%. This showed that it is a vital variable that affects economic growth now and would also affect it in the next period.

The coefficient of log of petroleum profit tax (LPPT) was 0.057789 proved that it also has a short- run time period in having a long-run effect on the dependent variable log of FS (LFS) both now and in the next period. The short-run speed of adjustment was approximately 6%. It also proved that it is an important variable that would impact the dependent variable both now and the next period in the long run. Finally, the coefficient of log of value added tax (LVAT) was −0.157526 and also proved that it also has a short-run time period in having a long-run effect on the dependent variable LFS both now and in the next period. The short-run speed of adjustment was approximately 16%. This also proved that it is an important variable that would impact the dependent variable both now and the next period in the long run.

Table 5. Vector error correction model result

Error Correction:	**D(LFS)**	**D(LCIT)**	**D(LPPT)**	**D(LVAT)**
CointEq1	−0.746375	−0.140119	0.057789	0.157526
	(0.03347)	(1.25701)	(3.31315)	(12.5702)
	[−5.59189]	[−0.90701]	[1.19457]	[0.45007]
D(LFS(−1))	0.214624	0.813967	6.680063	−13.99587
	(0.15368)	(1.44733)	(3.81478)	(14.4734)
	[1.39654]	[0.56239]	[1.75110]	[−0.96701]
D(LCIT(−1))	−0.078580	−0.375751	−0.301250	2.304278
	(0.03888)	(0.36616)	(0.96509)	(3.66159)
	[−2.02110]	[−1.02621]	[−0.31215]	[0.62931]
D(LPPT(−1))	0.016165	0.136293	−0.114494	−2.352489
	(0.00967)	(0.09104)	(0.23995)	(0.91039)
	[1.67218]	[1.49710]	[−0.47715]	[−2.58405]
D(LVAT(−1))	0.004902	−0.002705	−0.021427	−0.558952
	(0.00200)	(0.01884)	(0.04965)	(0.18836)
	[2.45095]	[−0.14363]	[−0.43160]	[−2.96749]
C	0.051113	0.168538	−0.209309	0.237088
	(0.00936)	(0.08812)	(0.23225)	(0.88116)

	[5.46286]	[1.91269]	[−0.90123]	[0.26906]
R-squared	0.809754	0.223859	0.226629	0.677085
Adj. R-squared	0.741809	−0.053334	−0.049575	0.561758
Sum sq. resids	0.005121	0.454227	3.155549	45.42333
S.E. equation	0.019126	0.180124	0.474759	1.801256
F-statistic	11.91779	0.807591	0.820514	5.871002
Log likelihood	54.32178	9.470144	−9.913073	−36.58171
Akaike AIC	−4.832,178	−0.347014	1.591307	4.258,171
Schwarz SC	−4.533459	−0.048295	1.890027	4.556890
Mean dependent	0.049950	0.161317	0.079803	−0.189534
S.D. dependent	0.037641	0.175505	0.463411	2.720936
Determinant resid covariance (dof adj.)		1.28E-06		
Determinant resid covariance		3.08E-07		
Log likelihood		36.42482		
Akaike information criterion		−0.842,482		
Schwarz criterion		0.551543		

Source: E views output

6.6 VECM Probability table

Table 6 displayed how quickly the independ variables changed how the dependent varial changed both now and over the following period. T table confirmed that there was a long-run relationsl between the dependent variable and all ı independent variables. The following phase involv determining which variable was significant and whi was not. This was investigated using the VEC probability table. A variable was significant if it falls below the 5% level of significance (0.05).

Using Table 6, C(2) represented the dependent variable. C(3) represented log of company income tax (LCIT) and its probability value of 0.0451 was statistically significant at 5% level of significance to prove that the variable was significant in impacting the dependent variable LFS

Table 6. VECM Probability Table

	Coefficient	Std. Error	t-Statistic	Prob.
C(1)	−0.746375	0.133475	−5.591889	0.0000
C(2)	0.214624	0.153683	1.396537	0.0381
C(3)	−0.078580	0.038880	−2.021104	0.0451

C(4)	0.016165	0.009667	1.672178	0.0000
C(5)	0.004902	0.002000	2.450949	0.0174
Determinant residual covariance		3.08E-07		

The coefficient value was −0.078580 and carries negative sign to show that the significance was negative one. Summarily, LCIT has a negative a significant relationship with LFS. C(4) represent LPPT and its significant figure was 0.0000 which w significant at 5% level of significance. Also, t coefficient value of 0.016165 carries a positive si Therefore, LPPT was positively significant impacting LFS. C(5) measured LVAT and significant value of 0.0174 was significant at 5% le of significance. Its coefficient value of 0.0049 carries a positive sign also. Hence, LVAT w positively significant in impacting LFS. C represented the error term.

7 Hypothesis Testing

For this study, three hypotheses were formulated for testing to analyse the effect of tax revenue on fiscal sustainability. To test the hypothesis, Vector Error Correction model was used to calculate the overall dependence of the dependent variable on the independent variable. The outcome of the vector error correction analysis will be used to test the hypothesis in this study. For the purpose of this study, a hypothesis is accepted at a 5% level of significance. A null hypothesis would only be rejected in Favor of the alternate if the P-value (probability value) is less than 0.05. A p-value greater than 0.05 will see the null hypothesis is accepted.

Table 7. Summary of hypothesis result

Null Hypothesis	Indicators	P value	T- value	Decision
H_01	PPT	0.0000	1.672	Reject
H_02	CIT	0.0451	−2.021	Reject
H_03	VAT	0.0174	2.450	Reject

The results of the hypothesis tested are as follows;

H_{01}: Petroleum profit tax has no significant effect fiscal sustainability in Nigeria

Using the probability table in Table 7, petrole profit tax was positively significant in impacting L with significant value of 0.0000. Therefore, based this finding, the null hypothesis was rejected and it accepted that petroleum profit tax has a signific effect on fiscal sustainability. Also based on t probability table in Table 7, company income tax w negatively significant in impacting LFS w significant value of 0.04511. Hence, the n hypothesis was rejected and we fail to reject t hypothesis that company income tax does ha significant effect on fiscal sustainability.

H_{02}: Company income tax has no significant effect fiscal sustainability in Nigeria

Based on the probability table in Table 7, compa income tax was negatively significant in impacti LFS with significant value of 0.04511. Hence, the null hypothesis was rejected and it is agreed that company income tax does have significant effect on fiscal sustainability in Nigeria.

H_{03}: Value added tax has no significant effect on fiscal sustainability in Nigeria

Moreover, evaluating the probability table in Table 7, value added tax was positively significant in impacting LFS with significant value of 0.0174. This means that the null hypothesis is rejected and it is accepted that value added tax have a significant impact on fiscal sustainability in Nigeria.

CONCLUSION

Tax revenue (measured by value added tax, company income tax, and petroleum profit tax were used as independent variables) against the dependent variable fiscal sustainability (FS). The findings revealed that all the three variables significantly influence fiscal sustainability, although company income tax w negatively significant; petroleum profit tax and val added tax were positively significant. Based on t finding, the study concludes that the relevance of t revenue to an improved Nigerian economy cannot over-emphasized. Therefore, tax revenue is an aven for the government to source for funds to use a improve the workings of the economy and this wo lead to fiscal sustainability. The findings from t study supported the Laffer curve theory. The Laf curve theory simply revealed that any change in t rate will have economic effect in areas of work, outp employment, and in consequence the tax base contributing to expand the activities throu incentives. The findings showed that tax revenu variables of company tax, petroleum tax and val added tax have significant impact on fis sustainability as a whole. Hence, any change in any the variable would impact the FS directly. T findings from the study also supported the endogen growth model which explained that fiscal sustenan and responsibility is influenced by internal fact rather than external factors. Tax was one of t internal factors as it is generated internally through t workings of human capital. The findings reveal t revenue to be significant in impacting fis sustainability, hence, supporting the theory.

RECOMMENDATIONS AND SUGGESTIONS FOR FURTHER STUDIES

Based on the findings, the study recommends trainings, the papers, and workshops should be organized by government tax agencies to the Nigerian public, students in higher institutions, and companies on the importance and benefits of tax revenue to the economy. There should be a more effective supervision of the tax revenue by the tax regulatory authorities. This would improve the safety and security of the purposes of tax revenue in Nigeria. Company income tax was negatively significant to the fiscal sustainability. Hence, tax authorities should encourage companies to pay tax so as to improve the growth of the economy which the companies would benefit from. Tax holidays or incentives should be given to companies and institutions and individuals who have been compliant to tax payment. This would encourage such institutions to keep paying tax as at when due. The Nigerian government should judiciously use the funds for the tax to improve the capital and recurrent expenditures and also improve on infrastructures. This would encourage the populace to benefit from the tax paid and continue to pay it. This study contributes to tax and fiscal sustainability literature by proving the importance of company income tax, petroleum profit tax, and value added tax to the tax sustainability. Further researchers can examine other tax revenues like the import duties and excise duties and how they impact Nigerian economic growth. The study enjoins other researchers to consider studying other nations in African continent for a comparative result.

REFERENCES

Adewale, K. A., & Adetiloye, K. A. (2021). Taxati and fiscal sustainability in Lagos Sta Nigeria. *International Journal of Advanc Research in Economics and Accounti* 3(1), 1-17.

Adewuyi, O. O. (2018). Tax policy and econon development: A case study of Oyo Sta Nigeria. European Scientific Journal, 14(1 111-125.

Akinboade, O. A., & Fadare, S. O. (2017). T policy and fiscal sustainability in Niger Journal of Policy Modeling, 39(5), 819-84

Akinola, G. A., & Adedokun, M. O. (2019). Taxati sustainable development and resou mobilization in Nigeria: Issues a challenges. *Research Journal of Finan and Accounting*, 10(11), 64-72.

Chibi, A., Chekouri, S. M., & Benbouziane, (2019). The dynamics of fiscal policy Algeria: sustainability and structural change. *Journal of Economic Structures*, 8(28), 1-27. doi.org/10.1186/s40008-019-0161-3.

Edmiston, K. D., Mudd, S., & Valav, S. (2003). Tax incentives and FDI: Evidence from state tax abatement programs. *The Journal of Regional Analysis & Policy*, 33(1), 1-17.

Idowu, H., Adeniji, O., & Ajemunigbohun, S. S. (2018). Tax policy and income inequality in Nigeria. *Research Journal of Finance and Accounting*, 9(17), 22-28.

Kaur, B., Mukhejeer, A., & Ekka, A. P. (2018). Debt sustainability in India: an assessment. *Indian Economic Review*, 53, 93-129

Kojo, N. C. (2010). Diamonds are not forever: Botswana medium-term fiscal sustainability. The World Bank Policy Research Working Paper No. 5480

Laffer, A. B. (2004). *The Laffer curve: past, present and future*. The Heritage Foundation.

Ndikom, O. (2020). Exploring the Nigerian t system in light of external debt and reven mobilization. *Afro-Asian Journal of Finar and Accounting*, 10(2), 132-147.

Ogunlana, A. O., & Afolabi, G. A. (2019). Taxati and financial sustainability of Nigeri states: An empirical investigation. *Journ of Economics and Sustainable Developme* 10(8), 115-122.

Oseni, S. O., & Okwu, A. T. (2017). Tax policy a sustainable development in Nigeria: overview. *International Journal of Busin and Management Invention*, 6(10), 56-61.

Okwu, A. T. (2018). The role of tax policy achieving environmental sustainability Nigeria. *International Journal of Busin and Management Invention*, 7(4), 34-43.

Oyeleke, O. J., & Ajilore, O. T. (2014). Analysis fiscal deficit sustainability in Nige economy: an error correction approach. *Asian Economic and Financial Review*, 4(2), 199-210

Presidential Committee on National Tax Policy. (2008). National Tax Policy (2nd ed.). Federal Inland Revenue Service.

PWC. (2020). Nigeria Tax Guide 2020. PricewaterhouseCoopers.

Sudharshan, C., Martin, B., Anca, P., & Ionut, D. (2012). The challenges to long run fiscal sustainability in Romaine. The World Bank Policy Research Working Paper, No. 5927

Taiwo, K. A., & Oyinbo, O. (2018). Fiscal policy and economic development in Nigeria: An empirical analysis. *Journal of Accounting and Taxation*, 10(2), 20-31.